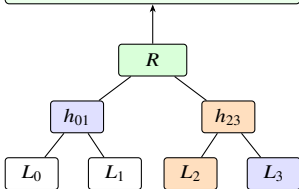


on-chain: `registerBatch( $R, n, cid, \dots$ )`



$$L_i = H(H(i, \text{addr}(sk_i)))$$

proof of unit 2: $\pi_2 = (L_3, h_{01})$

Label of unit 2 (leaf L_2)



public QR
/u/c/b/2



secret QR
satk2:c:b:2:sk2

scratched off after purchase

Buyer app scans the secret QR and signs with the unit key

$\sigma = \text{Sign}_{sk_2}(\text{EIP-712 ConsumeAuthorization}$

$(b, 2, \text{consumer}, \text{deadline}))$; the relay attaches π_2 and pays the gas.

consume(req, π_2 , σ) on-chain:

- 1) $\text{now} \leq \text{deadline}$
- 2) $\text{MerkleVerify}(\pi_2, R, L_2)$, L_2 rebuilt from $(2, \text{unitKey})$
- 3) lot not recalled; segment holding index 2 at the pharmacy; bit 2 unset
- 4) $\text{ecrecover}(\sigma) = \text{unitKey} \Rightarrow$ set bit 2; a second consume reverts